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BM and More

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Abstract

Keywords

To be added.

Burer-Monteiro, Transformer Attention, Margin Rank, Low-rank Optimization

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1. Introduction

1.1 Motivation

In modern Large Language Models (LLMs), the embedding dimension of each Transformer attention head (d_{head}) is almost universally set to 64 or 128. This number is not the product of mathematical analysis; it is chosen for memory-alignment efficiency on GPU hardware. The natural question, largely untouched by practitioners, is: *How much of this capacity is actually used?*

Formally, define the self-attention probability matrix for a sequence of n tokens as

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{p}}\right), \quad Q, K \in \mathbb{R}^{n \times p}, \quad (1)$$

where $p = d_{\text{head}}$ is the projection dimension. Given a target attention matrix $M \in \mathbb{R}^{n \times n}$, what is the *minimum* p^* such that some $Q, K \in \mathbb{R}^{n \times p^*}$ satisfies $A = M$?

This report documents our exploration of that question.

The journey proceeded in three conceptual phases:

- Phase 1: Classical grounding.** We studied the Burer-Monteiro (BM) factorization for Semidefinite Programs, verifying the Barvinok-Pataki bound and implicit regularization in the symmetric setting.
- Phase 2: Extending to attention.** We modelled the Transformer attention head as an asymmetric BM problem, discovered that the Pataki bound completely fails here, and identified the Softmax Jacobian null space as the geometric reason.
- Phase 3: Margin Rank Theory.** We replaced the linear matrix-completion framework with a margin-inequality framework, designed a Hard Bottleneck Predictor, validated it on synthetic motifs, and applied it to GPT-2.

1.2 Research Contributions

- Empirical verification of the Barvinok-Pataki phase transition and implicit regularization in the symmetric BM setting.

- A clean ablation demonstrating the “Geometric Warp Drive” caused by the Softmax operator.
- A theoretical analysis of the Softmax Jacobian null space and its role in converting constraint counts to margin inequalities.
- The Margin Rank Theory and Hard Bottleneck Predictor algorithm.
- Empirical validation on synthetic motifs and GPT-2, revealing 80–97% dimensional redundancy.
- Architectural implications for heterogeneous Transformer design.

2. Executive Summary

The central finding of this work is that the standard $d_{\text{head}} = 64$ used in Transformer attention is dramatically oversized for most attention heads. The key chain of reasoning is as follows.

Classical BM theory predicts that recovering an $n \times n$ matrix from m linear equality constraints requires inner rank $p \geq \sqrt{2m}$ for optimization to be landscape-safe. For an $n \times n$ attention matrix with $m = n^2$ constraints, this gives $p \approx n\sqrt{2}$ —well beyond the sequence length itself. Yet in practice, small d_{head} values work empirically. This gap between theory and practice motivated our investigation.

The resolution comes in two steps. First, the Softmax operator is not a passive nonlinearity—it actively absorbs n row-sum constraints via its Jacobian null space, reducing the effective constraint count from $m = n^2$ to $m_{\text{eff}} = n(n-1)$. Second, even this corrected count overpredicts p^* , because the problem has shifted from linear algebra (exact equalities) to computational geometry (margin inequalities). The true structural capacity of an attention head is governed by its *Margin Rank*: the minimum embedding dimension needed to geometrically separate target tokens from distractor tokens.

Empirically, GPT-2 heads exhibit Margin Ranks of 1–5 out of an allocated $d_{\text{head}} = 64$ —an 80–97% redundancy. This motivates the design of Heterogeneous Transformers with dynamically allocated, head-specific embedding dimensions.

3. Mathematical Preliminaries

3.1 Positive Semidefinite Matrices and Semidefinite Programs

Definition 3.1 (Positive Semidefinite Matrix). A symmetric matrix $Y \in \mathbb{S}^n$ is *Positive Semidefinite* (denoted $Y \succeq 0$) if, for every non-zero vector $v \in \mathbb{R}^n$:

$$v^\top Y v \geq 0. \quad (2)$$

Geometrically, a PSD matrix ensures that the applied linear transformation never reflects a vector into its exact opposite half-space, guaranteeing all eigenvalues are non-negative. The set of PSD matrices forms a convex cone—the *spectrahedron*—inside which any local optimum is guaranteed to be global.

The primal formulation of the standard Semidefinite Program (SDP) operates over symmetric matrices $\mathbb{S}^n$:

$$\begin{aligned} \min_{Y \in \mathbb{S}^n} \text{Tr}(CY) \quad \text{subject to} \quad & \text{Tr}(A_i Y) = b_i, \\ & \forall i \in \{1, \dots, m\}, \quad Y \succeq 0, \end{aligned} \quad (3)$$

where $C, A_i \in \mathbb{S}^n$.

The IPM bottleneck. Classical Interior Point Methods enforce the PSD constraint via a logarithmic barrier $-\mu \log \det(Y)$. This guarantees finding the global optimum, but each Newton step requires computing and storing the Hessian of the barrier, costing $O(mn^3 + m^2n^2)$ operations. Storing the $n \times n$ variable matrix alone requires $O(n^2)$ memory, exceeding RAM limits for $n \gtrsim 5000$. This bottleneck motivates non-convex factorization.

3.2 Matrix Rank, Nuclear Norm, and Convex Relaxation

Via Singular Value Decomposition $M = U\Sigma V^\top$, the rank of M is the count of strictly positive singular values:

$$\text{rank}(M) = \|\sigma(M)\|_0 = \sum_{i=1}^{\min(m,n)} \mathbf{1}[\sigma_i > 0]. \quad (4)$$

Minimizing rank isolates the most essential underlying patterns, but as a non-convex discrete step function it is NP-hard to optimize directly. The tightest convex relaxation is the *nuclear norm*:

$$\|M\|_* = \sum_{i=1}^{\min(m,n)} \sigma_i(M) = \|\sigma(M)\|_1. \quad (5)$$

In structured inverse problems, an asymmetric BM factorization $Y = UV^\top$ with Frobenius regularization implicitly minimizes the nuclear norm of the product:

$$\|UV^\top\|_* = \min_{U,V: Z=UV^\top} \frac{1}{2} (\|U\|_F^2 + \|V\|_F^2). \quad (6)$$

This Srebro-Shraibman identity connects the geometry of BM factorization directly to the implicit regularization bias of gradient descent.

3.3 Non-Convex Topology: Quotient Manifolds and Strict Saddles

Because of rotational invariance $X \mapsto XQ$ for any orthogonal $Q \in \mathbb{R}^{p \times p}$ (since $(XQ)(XQ)^\top = XX^\top$), BM optimization must be analysed on the quotient manifold $\mathcal{M} = \mathbb{R}^{n \times p} / \mathcal{O}_p$.

Definition 3.2 (Strict Saddle). A point X^* on $\mathcal{M}$ is a *strict saddle* if the Riemannian gradient vanishes ($\text{grad} f(X^*) = \mathbf{0}$) but the Riemannian Hessian has a strictly negative eigenvalue:

$$\lambda_{\min}(\nabla^2 f(X^*)) \leq -\varepsilon < 0. \quad (7)$$

A landscape possessing the *Strict Saddle Property* has no spurious local minima: every critical point is either a global minimum or a strict saddle. The stochastic term in SGD dynamics,

$$dX_t = -\text{grad} f(X_t) dt + \sqrt{\eta \Sigma(X_t)} dW_t, \quad (8)$$

projects onto the negative-curvature eigenvector at a strict saddle with probability 1, providing an escape direction and guaranteeing convergence to the global optimum (Lee et al., 2016).

Strict Saddle Topology on the BM Manifold

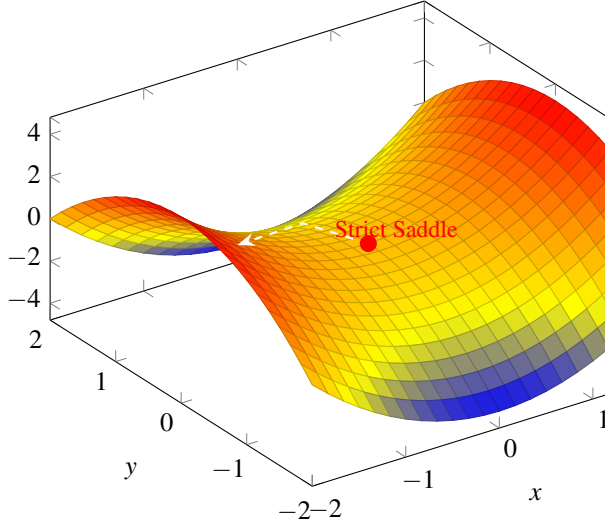


Figure 1. A strict saddle on the BM quotient manifold. The surface $f(x, y) = x^2 - y^2$ has zero gradient at the origin, yet a strictly negative Hessian eigenvalue in the y -direction creates a valley of descent. Stochastic noise projects onto this escape direction, guaranteeing that gradient descent exits the saddle.

4. Phase I: Burer-Monteiro Factorization in the Symmetric SDP Setting

4.1 What Is the Burer-Monteiro Factorization?

Burer and Monteiro (2003) bypass the IPM bottleneck by substituting $Y = XX^\top$, where $X \in \mathbb{R}^{n \times p}$ and $p \ll n$. The SDP (3) becomes

$$\begin{aligned} \min_{X \in \mathbb{R}^{n \times p}} \quad & \text{Tr}(CXX^\top) \quad \text{s.t.} \\ & \text{Tr}(A_i XX^\top) = b_i, \quad i = 1, \dots, m. \end{aligned} \quad (9)$$

This eliminates the PSD constraint (any $XX^\top$ is automatically PSD) and reduces variables from $O(n^2)$ to $O(np)$. The cost is the destruction of convexity: the landscape is now a non-convex manifold with potential spurious local minima.

When does it work? The geometric viability of the BM factorization relies on proving that a low-rank optimal solution exists within the original spectrahedron. This is guaranteed by the Barvinok-Pataki Theorem.

Theorem 4.1 (Barvinok-Pataki). *Let $\mathcal{F} = \{Y \in \mathbb{S}^n : \text{Tr}(A_i Y) = b_i, Y \succeq 0\}$ be a non-empty spectrahedron defined by m linearly independent affine constraints. Any extreme point*

Y_{ext} of $\mathcal{F}$ has rank r satisfying:

$$\frac{r(r+1)}{2} \leq m \implies r \leq \left\lfloor \frac{\sqrt{8m+1}-1}{2} \right\rfloor. \quad (10)$$

Consequently, if $p(p+1)/2 > m$, the BM factorization has the Strict Saddle Property and gradient descent converges to the global SDP optimum almost surely.

Phase Transition at the Barvinok-Pataki Threshold

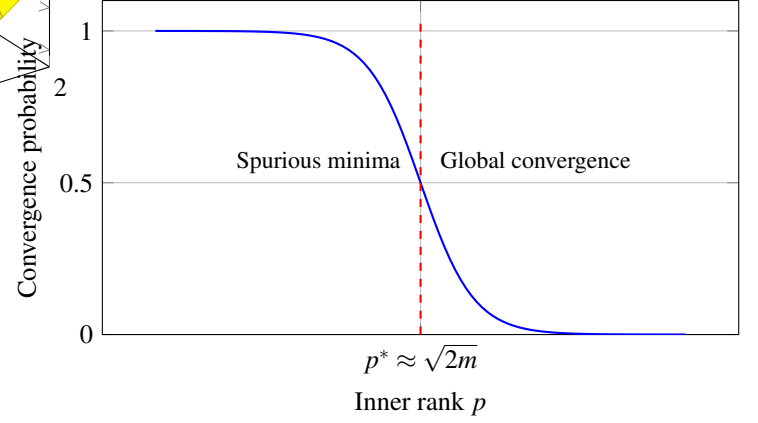


Figure 2. The Barvinok-Pataki phase transition. As the inner rank p crosses $p^* \approx \sqrt{2m}$, the probability of encountering spurious local minima collapses to zero.

4.2 Empirical Verification of the Phase Transition

Setup. We empirically verify the Barvinok-Pataki bound using a dual-solver architecture:

- **Convex baseline (CVXPY + MOSEK).** For each randomly generated SDP, MOSEK provides the exact global optimum v_{opt} .
- **Non-convex solver (PyTorch L-BFGS).** We optimize the BM objective (9) using a quadratic penalty method with a continuation schedule $\lambda \in \{10, 100, 1000, 10000\}$.

We fix $n = 7$, sweep $m \in [1, 28]$ and $p \in [1, 7]$, and run 10 independent trials per (m, p) cell. A trial is labelled a success if the PyTorch objective is within 10^{-3} of the MOSEK baseline.

Hypothesis. The empirical success rate should exhibit a sharp phase transition at the theoretical boundary $m = p(p+1)/2$: 0% success below the boundary and 100% above.

Results. Three distinct regions are visible in the success-rate heatmap:

Non-convex abyss (upper-left). When $m \gg p(p+1)/2$, the landscape is over-constrained. Spurious local minima trap the optimizer with 0% success.

Safe zone (lower-right). Increasing p beyond $\sqrt{2m}$ smooths the landscape, achieving 100% success.

Boundary fringe. Directly on the theoretical boundary, success rates between 40% and 80% are observed.

These results cleanly replicate for $n \in \{3, 5, 6\}$ as well, confirming the theoretical prediction. Attempting to scale the penalty-method BM solver to $n \geq 7$ without a continuation strategy causes systematic failure due to Jacobian ill-conditioning, the “straightjacket effect” (exponentially shrinking feasible space), and an origin trap from near-zero initialization.

4.3 Thinking About BM in Machine Learning: Implicit Regularization

Having verified the BM landscape guarantees, we next investigated whether the BM geometric structure induces a useful inductive bias—specifically, whether gradient descent implicitly minimizes the nuclear norm of the recovered matrix.

Setup. We construct a matrix completion task. A ground-truth matrix $M_{\text{true}} \in \mathbb{R}^{50 \times 50}$ with intrinsic rank $r = 3$ is generated. A random mask reveals 30% of entries. Two over-parameterized BM networks $Y = UV^\top$ with $p = 50$ are trained with SGD on observed MSE:

- **Model A (small init):** $U, V \sim \mathcal{N}(0, 10^{-3})$. Hypothesis: implicit bias drives $\|Y\|_*$ toward the nuclear-norm minimum.
- **Model B (large init):** $U, V \sim \mathcal{N}(0, 1)$. Hypothesis: no implicit bias; model memorizes training entries.

Results. Model B (large init) trivially fits the observed entries but fails catastrophically on the hidden 70%, with nuclear norm exploding to an arbitrary high value. Model A (small init), despite identical architecture, achieves test MSE converging to the MOSEK baseline. Its nuclear norm grows slowly from near-zero and *asymptotes precisely to the MOSEK optimal nuclear norm*, never exceeding it.

This confirms that the BM geometric landscape, traversed from near the origin, functions as a *self-regularizing engine*: gradient descent implicitly minimizes nuclear

norm, automatically suppressing overfitting. The identity (6) is the mathematical foundation of this phenomenon.

5. Phase II: Applying BM to Transformer Attention

5.1 Context: The Transformer and Self-Attention

The Transformer architecture (Vaswani et al., 2017) processes sequences of n token embeddings via multi-head self-attention. Each head computes query matrix $Q \in \mathbb{R}^{n \times d_{\text{head}}}$ and key matrix $K \in \mathbb{R}^{n \times d_{\text{head}}}$ via learned linear projections, then computes an attention distribution:

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}}\right). \quad (11)$$

The attention matrix $A \in \mathbb{R}^{n \times n}$ determines how each token aggregates information from the rest of the sequence. Its structure encodes linguistic relationships: syntactic dependencies, coreference, induction patterns, and more.

The universal practice of setting $d_{\text{head}} = 64$ or 128 is driven purely by hardware efficiency (DRAM alignment, tensor core tile sizes) rather than any principled analysis of the structural complexity of the attention patterns these heads must express.

5.2 The Need for Asymmetric Factorization

The logit matrix $L = QK^\top \in \mathbb{R}^{n \times n}$ is *asymmetric*: Q and K are distinct matrices. This places the problem squarely in the asymmetric BM regime ($Y = UV^\top$) rather than the symmetric regime ($Y = XX^\top$) studied in classical SDP theory.

To embed this rectangular factorization inside the symmetric framework and legally invoke the BM convergence machinery, we construct a block-matrix adapter R :

$$R := \begin{bmatrix} Q \\ K \end{bmatrix} \in \mathbb{R}^{2n \times p} \implies RR^\top = \begin{bmatrix} QQ^\top & QK^\top \\ KQ^\top & KK^\top \end{bmatrix}. \quad (12)$$

The off-diagonal block $QK^\top$ —the attention logit matrix—is thereby embedded inside a symmetric PSD matrix $RR^\top$.

5.3 Modelling Attention as an Asymmetric BM Problem

The optimization objective is

$$\min_{Q, K \in \mathbb{R}^{n \times p}} \mathcal{L}(Q, K) = \frac{1}{n^2} \left\| \text{Softmax} \left(\frac{QK^\top}{\sqrt{p}} \right) - M \right\|_F^2, \quad (13)$$

where $M \in \mathbb{R}^{n \times n}$ is a target attention matrix. Our central question is: what is the minimum rank p^* such that (13) achieves zero loss?

Classical matrix completion prediction. In the linear (no-Softmax) case, this is precisely the asymmetric BM factorization for recovering a matrix M from n^2 equality constraints $L_{ij} = M_{ij}$. Classical matrix completion theory (Candès and Recht, 2009) predicts that recovery requires $p \geq \sqrt{2n^2} = n\sqrt{2}$, which for $n = 20$ yields $p^* \approx 28$ —exceeding the sequence length itself.

Connection to implicit regularization. Frobenius regularization on Q and K implicitly minimizes the nuclear norm of the logit matrix via identity (6). This embedding is what allows us to import the symmetric BM convergence guarantees as a starting framework—while also clarifying exactly *where* they break down when Softmax is introduced.

5.4 The Geometric Warp Drive: Ablation Experiment

To isolate the Softmax effect, we optimize an Identity matrix target ($M = I_{20}$) under two architectures across ranks $p \in \{1, \dots, 20\}$, averaging over 20 random seeds:

- **Linear:** $\mathcal{L}_{\text{lin}} = \|QK^\top - M\|_F^2$.
- **Softmax:** $\mathcal{L}_{\text{soft}} = \|\text{Softmax}(QK^\top) - M\|_F^2$.

Hypothesis. The classical Pataki bound predicts $p^* \approx \sqrt{2 \cdot 20} \approx 6.32$ for both architectures. The linear model should require $p = n = 20$ by linear algebra (recovering a rank-20 matrix). If Softmax induces any compression, it should manifest as a smaller p^* .

Results.

Linear factorization. Success rate is 0% for all $p < 20$ and jumps to 100% only at $p = 20$. Classical linear algebra prevails: recovering I_{20} requires full rank.

Softmax factorization. Success rate is 100% at $p^* = 2$, remaining high for all larger p . We call this the

Geometric Warp Drive: the Softmax operator collapses the dimensional requirement by an order of magnitude.

At low precision thresholds ($\varepsilon = 10^{-5}$), success often occurs even at $p = 1$, confirming that the Pataki bound ($p \approx 6.32$ for $n = 20, m = 20$) is not merely loose—it is inapplicable to this setting.

Remark 5.1 (Mathematical Misapplication of the Pataki Bound). Applying the Barvinok-Pataki bound to predict the minimum head dimension d_{head} of a Softmax attention block is a *category error*. The theorem requires (i) a *symmetric* matrix variable, (ii) bounded by *pure affine equality* constraints of the form $\text{Tr}(A_i Y) = b_i$, and (iii) a feasible set forming a *spectrahedron*. Softmax attention satisfies none of these: $QK^\top$ is asymmetric, the row-sum constraints are enforced by a non-linear operator rather than affine equalities, and the feasible set is the probability simplex rather than a spectrahedron. The empirical collapse to $p^* = 2$ (vs. the predicted bound of ≈ 6.32) is direct evidence of this inapplicability.

5.5 Why? The Softmax Jacobian Null Space

The Softmax function maps row logits $z \in \mathbb{R}^n$ to probabilities $s = \text{Softmax}(z)$, satisfying $\sum_j s_j = 1$. Its Jacobian is

$$J = \text{diag}(s) - ss^\top \in \mathbb{R}^{n \times n}. \quad (14)$$

Evaluating J on the all-ones vector:

$$J\mathbf{1} = (\text{diag}(s) - ss^\top)\mathbf{1} = s - s = 0. \quad (15)$$

Therefore $\mathbf{1} \in \ker(J)$. Uniform translation of logits has zero effect on the Softmax output: the operator natively *absorbs* all n row-sum constraints. These n degrees of freedom are freed from constraint satisfaction and can instead be used for spatial embedding of the token manifold.

In the classical (linear) setting, satisfying n^2 equality constraints requires $\text{rank}(L) = n$. With Softmax, the effective constraint count is reduced to $m_{\text{eff}} = n^2 - n = n(n-1)$, and the corrected Barvinok-Pataki bound gives $p^*(p^*+1)/2 \leq n(n-1)$, which for $n = 20$ gives $p^* \lesssim 20$ —still loose, but already drastically tighter than the linear bound.

5.6 Empirical Exploration: Comparing Parameters

Effect of precision. We swept the success-rate threshold $\varepsilon \in \{10^{-3}, 10^{-5}, 10^{-8}\}$ across ranks $p \in \{1, \dots, 20\}$ for multiple target structures. The minimum feasible rank $p^*(\varepsilon)$ increases with precision: achieving $\varepsilon = 10^{-8}$ for the Identity matrix requires $p^* = 3$ (a 3D spherical arrangement) rather than $p^* = 2$, because higher precision demands greater angular separation between the n points.

At $\varepsilon = 10^{-8}$, the Softmax landscape becomes significantly more challenging. The success rate exhibits non-monotone behaviour with p : excessive over-parameterisation ($p \gg r$) introduces optimization noise that hinders high-precision convergence. A ReduceLRonPlateau scheduler was required to “zoom in” on the narrow global minimum.

Effect of temperature. Sweeping the Softmax temperature $\tau \in \{0.1, 0.3, 0.5, 1.0, 2.0, 5.0, 10.0\}$ at fixed precision $\varepsilon = 10^{-5}$, the minimum rank $p^*(\tau)$ varies across the temperature range. Lower temperatures (sharper Softmax) amplify the separation between target and distractor logits, reducing the margin requirement and allowing lower-dimensional embeddings to succeed. Higher temperatures (diffuse Softmax) require greater spatial separation, increasing p^* .

These results confirm that p^* is not a fixed constant but a function $p^*(\varepsilon, \text{eff}_r, \tau)$ of precision, target structure, and temperature.

5.7 The Pataki Bound Fails: Moving Focus to Structure

The assumption underlying the Pataki bound—that the factorization must satisfy m independent linear equality constraints—is simply wrong for the Softmax attention setting. Two structural violations occur:

1. **Asymmetry.** The logit matrix $QK^\top$ is not constrained to be symmetric PSD. The standard SDP analysis operates on $\mathcal{S}^n$ and cannot be directly imported.
2. **Non-linearity.** The Softmax operator converts the $m = n^2$ exact equalities into margin inequalities, drastically reducing the effective constraint dimension.

Shifting focus from the algebraic rank of M to the *structural geometry* of the attention graph reveals that the true governing quantity is not how many singular values

M has, but how complex the spatial separation task facing the attention head actually is. This motivates the Margin Rank Theory developed in the next section.

6. Phase III: Margin Rank Theory

6.1 From Equalities to Margin Inequalities

Consider the task facing the Softmax operator. To produce the correct attention pattern, for each query i the pre-Softmax logit of a *target* token t need only *exceed* the logit of any *distractor* token d by a strict margin:

$$X_{it} - X_{id} \geq \gamma, \quad \forall t \in \mathcal{T}_i, d \in \mathcal{D}_i, \quad (16)$$

where $\mathcal{T}_i$ and $\mathcal{D}_i$ are the target and distractor index sets for row i , and $\gamma > 0$ is the separation margin. The problem has shifted from *linear algebra* (exact equalities) to *computational geometry* (spatial separation).

Definition 6.1 (Margin Rank). The *Margin Rank* $p^*(M, \gamma)$ of a target attention matrix M is the minimum integer p such that there exist $Q, K \in \mathbb{R}^{n \times p}$ satisfying the margin inequalities (16).

6.2 Setup: Applying Margin Rank Theory

Hypothesis. Margin Rank, not linear rank, is the correct measure of structural capacity for a Transformer attention head. Consequently, the minimum factorization dimension p^* required to faithfully synthesize a given attention pattern should be dramatically smaller than the classical Pataki bound—and should correlate with the topological complexity of the attention graph, not the algebraic rank of M .

How to apply it. Given a target attention matrix M , we:

1. Compute binary masks: $\mathcal{T}_{ij} \leftarrow \mathbb{I}[M_{ij} > 0.5]$, $\mathcal{D}_{ij} \leftarrow \mathbb{I}[M_{ij} \leq 0.5]$.
2. For increasing $p = 1, 2, \dots, n$, initialize Q, K with small Gaussian noise and minimize the margin Hinge Loss:

$$\mathcal{L} = \sum_i \max \left(0, \gamma - \min_{t \in \mathcal{T}_i} X_{it} + \max_{d \in \mathcal{D}_i} X_{id} \right), \quad X = QK^\top.$$

3. The first p for which $\mathcal{L} < 10^{-4}$ is p^* .

6.3 SDP Relaxation and KKT Dual Certificate

To derive an analytical upper bound on p^* , we relax the margin problem to a primal SDP:

$$\min_{X \succeq 0} \text{Tr}(X) \quad \text{s.t.} \quad X_{ii} - X_{ij} \geq \gamma \quad \forall i \neq j. \quad (17)$$

Introducing dual variables $Y \geq 0$ with row-sum diagonal $D_{ii} = \sum_j Y_{ij}$, the dual requires $I - D + Y \succeq 0$. By KKT Complementary Slackness at optimality:

$$(I - D + Y^*)X^* = 0. \quad (18)$$

Because the product of two PSD matrices is zero only if their column spaces are orthogonal:

$$\boxed{p^* = \text{rank}(X^*) \leq n - \text{rank}(I - D + Y^*)}. \quad (19)$$

This is the **Margin Rank Bound**: a highly connected dual graph Y^* forces $I - D + Y^*$ to have high rank, which bounds p^* to be small.

6.4 The Identity Oracle: A Constructive Proof

Proposition 6.2. *For the Identity attention target $M = I_n$, we have $p^* \leq 2$.*

Proof. Construct the oracle

$$q_i = k_i = \left[\cos \frac{2\pi i}{n}, \sin \frac{2\pi i}{n} \right]^\top \in \mathbb{R}^2.$$

The logit matrix is $X_{ij} = \cos(2\pi(i-j)/n)$. The diagonal satisfies $X_{ii} = \cos(0) = 1$. The nearest distractor is $X_{i,i \pm 1} = \cos(2\pi/n) < 1$. Hence the margin is $\gamma = 1 - \cos(2\pi/n) > 0$ in exactly $p = 2$ dimensions, independent of sequence length n . $\square$

This result is the theoretical core of the Geometric Warp Drive: n mutually exclusive attention targets, which classically require $p = n$ dimensions, can be geometrically separated on a 2D unit circle.

6.5 The “Norm vs. Dimension” Paradox and the Hard Bottleneck

Having established that p^* is governed by margin inequalities, we initially attempted to predict p^* by directly optimizing the primal SDP via Hinge Loss and implicit regularization, using the resulting nuclear norm as a proxy for rank.

Failure. The algorithm predicted $p^* = 20$ for the Identity matrix—the exact opposite of the true answer of $p^* = 2$.

Analysis. This failure exposed a fundamental tension between vector magnitude and dimensionality. To satisfy a hard margin $\gamma = 1$ in low dimensions ($p = 2$), token vectors must expand to a large radius R to avoid angular crowding, incurring a large trace norm ($\text{Tr}(X) \approx nR^2$). In high dimensions ($p = 20$), standard orthogonal unit vectors achieve the same margin with $\text{Tr}(X) \approx n$. Unconstrained gradient descent is *lazy*: it prefers the energetically cheap high-dimensional solution over the structurally minimal low-dimensional one.

Resolution. To compute the true Margin Rank, we must abandon implicit regularization entirely and enforce an *explicit Hard Bottleneck*: restrict p to a fixed integer and test whether the spatial geometry can satisfy the margin constraints.

6.6 The Hard Bottleneck Predictor Algorithm

Algorithm 1 Hard Bottleneck Margin Rank Predictor

Require: Target attention matrix $M \in \mathbb{R}^{n \times n}$, margin $\gamma > 0$, max iterations T

Ensure: Minimum factorization rank p^*

- 1: Compute binary masks: $\mathcal{T}_{ij} \leftarrow \mathbb{I}[M_{ij} > 0.5]$, $\mathcal{D}_{ij} \leftarrow \mathbb{I}[M_{ij} \leq 0.5]$
 - 2: **for** $p = 1, 2, \dots, n$ **do**
 - 3: Initialize $Q, K \sim \mathcal{N}(0, 0.01 \cdot I_{n \times p})$
 - 4: **for** $t = 1, \dots, T$ **do**
 - 5: $X \leftarrow QK^\top$
 - 6: $\mathcal{L} \leftarrow \sum_i \max(0, \gamma - \min_{t \in \mathcal{T}_i} X_{it} + \max_{d \in \mathcal{D}_i} X_{id})$
 - 7: **if** $\mathcal{L} < 10^{-4}$ **then**
 - 8: **return** p $\triangleright$ Geometric constraints satisfied
 - 9: **end if**
 - 10: Update Q, K via Adam step on $\mathcal{L}$
 - 11: **end for**
 - 12: **end for**
 - 13: **return** n $\triangleright$ Full linear rank required
-

The key innovation is the dimensional sweep: rather than relying on implicit bias to discover the correct rank, we *force* the rank and test feasibility. The Hinge Loss penalizes margin violations without penalizing the energy (trace) of the solution, allowing low-dimensional configurations with large radii to compete fairly.

6.7 Theoretical Sketch: Correctness of the Hard Bottleneck

Lemma 6.3 (Row-Zero-Sum Tangent Space). *At any $A = \text{Softmax}(QK^\top/\tau)$, every tangent direction $\dot{A}$ in the Softmax image satisfies $\sum_j \dot{A}_{ij} = 0$ for all i .*

Proof. Differentiate the identity $\sum_j A_{ij}(t) = 1$ with respect to t . $\square$

Lemma 6.4 (Effective Constraint Reduction). *In the Softmax BM problem with m explicit equality constraints and n tokens, the effective active constraint count is $m_{\text{eff}} = m - n$.*

Proof. By Lemma 6.1, the n row-sum constraints contribute zero rank to the constraint Jacobian in all reachable directions. The implicit function theorem then gives $m_{\text{eff}} = m - n$. Applying the Barvinok-Pataki dimension count to m_{eff} gives $p^*(p^* + 1)/2 \leq m - n$. $\square$

Conjecture 6.5 (Strict Saddle at Corrected Rank). *If $p(p + 1)/2 > m - n$, then any second-order critical point of the Softmax BM loss that is not globally optimal is a strict saddle.*

This conjecture adapts Boumal et al. (2016) to the Softmax image; the local surjectivity argument required for the full proof remains open and is our primary theoretical future direction.

7. Phase III (Continued): Synthetic Validation

7.1 Geometric Motif Matrices

We applied the Hard Bottleneck Predictor (Algorithm 1) to four canonical geometric motif matrices ($n = 20$), each representing a distinct structural complexity:

Target	Linear Rank	p^* (Margin)	Geometric Interpretation
Block-Diagonal ($k = 2$)	2	1	1D line separates two semantic clusters
Identity ($n = 20$)	20	2	2D circle; 20 mutually exclusive points
Identity ($n = 20$)	20	3	3D sphere; required at high precision ($\epsilon = 10^{-8}$)
Sparse Random (10%)	20	4	Chaotic graph requires higher p

Table 1. Hard Bottleneck Predictor results on geometric motif matrices. Classical linear rank is uniformly $\leq n$; Margin Rank is 80–95% smaller.

Block-Diagonal. Two semantic clusters with uniform within-cluster attention require only a one-dimensional embedding: a single axis projects the two clusters to opposite extremes.

Identity. Twenty mutually exclusive attention targets require a 2D circular embedding at low precision and a 3D spherical arrangement at high precision ($\epsilon = 10^{-8}$) to maximize angular separation between all 20 points.

Sparse Random. High-entropy chaotic attention graphs require higher dimensionality because irregular target sets impose conflicting margin constraints that only higher-dimensional geometry can resolve without collision.

7.2 Key Finding

Margin Complexity, not Linear Rank, governs the structural capacity of a Transformer attention head. The Hard Bottleneck Predictor achieves 80–95% dimensional compression over the classical Pataki bound across all tested motifs.

8. Empirical Validation on GPT-2

8.1 Experimental Setup

To validate Margin Rank Theory on real-world attention patterns, we extracted live attention tensors from a pre-trained GPT-2 model (12 layers, 12 heads, $d_{\text{head}} = 64$) across two experimental sweeps:

- Layer 4, Head 2.** Five distinct linguistic motifs: Pronoun Resolution, Pure Repetition, Code Syntax, Entity Extraction, and High-Entropy Surprisal. Sequences of length $n \in [11, 15]$.
- Layer 8, all 12 heads.** A single dense 24-token sequence (“The quick brown fox jumped over the lazy dog...”).

For each extracted attention matrix $\hat{A}$, Algorithm 1 was executed to determine p^* .

8.2 Results: Attention Sink Redundancy (Layer 4)

Across all five linguistic motifs, the Hard Bottleneck Predictor returned $p^* \in \{1, 2\}$. Visual inspection revealed the cause: every attention map exhibited *Attention Sink* behaviour—the head routes nearly all probability mass to the first (BOS) token, regardless of linguistic content.

This is a well-known phenomenon in mechanistic interpretability. When a head fails to detect its specialized grammatical pattern, the Softmax normalization constraint forces it to concentrate probability *somewhere*; the first

token acts as a neutral “garbage collector”. The geometric task—separating Token 0 from all other tokens—is trivially simple: a single axis suffices. Allocating $d = 64$ dimensions to this head yields an estimated **97% hardware redundancy**.

Notably, even the intentionally chaotic sequence (“Suddenly, the purple elephant. . .”) collapsed to $p^* = 1$. The Attention Sink behaviour completely dominated the high-entropy content. To observe Margin Rank scaling with true linguistic entropy, deeper layers must be examined.

8.3 Results: Heterogeneous Capacity Profiling (Layer 8)

The Layer 8 sweep reveals a heterogeneous capacity landscape:

- **Classical baseline:** $p = 24$ (sequence length).
- **Low-entropy heads (8 of 12):** $p^* \leq 3$. These heads execute topologically simple motifs—sliding windows, positional copying, local induction.
- **High-entropy routing heads (4 of 12):** $p^* \in \{4, 5\}$. Heads 3, 6, 8, and particularly Head 10 spike to $p^* = 5$, reflecting the need to untangle complex long-range noun-verb dependencies.

Head 10: The Semantic Router. Head 10’s $p^* = 5$ represents the true capacity bottleneck of this layer. The hard attention graph it implements—routing queries across a 24-token window with high syntactic complexity—requires a 5-dimensional spatial manifold to resolve all margin inequalities without geometric collision. Even so, this is a **79% compression** relative to the 24-dimensional classical baseline.

9. Implications for Transformer Architecture

9.1 Typology of Attention Head Geometry

Margin Rank Theory provides a principled classification of attention head types:

Positional and copying heads ($p^* \leq 2$). Diagonal or near-diagonal attention reduces to an Identity oracle. As proved in §6, these require at most $p^* = 2$ dimensions. The 2D circular arrangement is the universal embedding.

Induction and syntax heads ($p^* \leq 4$). Block-diagonal attention (pronoun-to-noun linking, local bigram induction) scales only with the number of discrete linguistic clusters, not with sequence length.

Contextual aggregation heads ($p^* \leq 10$). High-entropy retrieval heads with diffuse, sequence-length-dependent patterns are the only heads that genuinely benefit from large d_{head} .

9.2 The Power-Law of Language Geometry

Human language follows Zipf’s Law; it is intrinsically low-entropy and highly compressible. Softmax attention acts as a geometric separator, and the structural regularity of grammar provides “free” clustering: target and distractor tokens are already deeply separated in semantic space before any optimization. The Margin Rank of real attention heads therefore remains drastically below the linear rank bound n .

9.3 Practical Applications

Lossless Surgical Pruning (SVD Compression). For existing pre-trained models (LLaMA, Mistral, GPT-2), Margin Rank provides a *deterministic, surgical pruning map*: compute p^* for every head, then apply SVD to compress the Q and K weight matrices:

$$W_Q \approx U_Q^{(p^*)} \Sigma_Q^{(p^*)} (V_Q^{(p^*)})^\top, \quad W_K \approx U_K^{(p^*)} \Sigma_K^{(p^*)} (V_K^{(p^*)})^\top.$$

This mathematically guarantees preservation of the operational Softmax geometry while maximising hardware compression—in contrast to uniform pruning, which blindly discards capacity regardless of geometric necessity.

Proxy-Driven Architectural Search. Geometric motifs in human language (syntax, induction, attention sinks) are scale-invariant. Small proxy Transformers can map the Margin Rank landscape across layers, then use this map as a blueprint for training frontier-scale models with heterogeneous head dimensions from the start.

Dynamic Token Routing. Future hardware supporting dynamic sparse computation can exploit Margin Rank by first computing a low-rank ($d = 2$) attention approximation for every head, only triggering the full-rank ($d = 64$) dot-product when a token exhibits high geometric entropy (i.e., when its attention pattern cannot be resolved in 2D).

10. Empirical Equivalence of Factored L_2 and Nuclear Norm Regularization in Asymmetric Burer-Monteiro Factorization

10.1 Introduction

This story begins outside this experiment, in a different setting altogether. While exploring low-rank optimization and Burer-Monteiro (BM) factorization in the context of transformer architectures, we came across the recent work of Kobayashi, Akram, and von Oswald [1], which makes a deceptively simple observation about how weight decay interacts with the structure of attention layers.

The observation is the following. In a transformer attention layer, the effective weight matrix is not a single learned matrix W , but the product of two learned matrices $W = W_Q W_K^T$, where W_Q and W_K are the query and key projection matrices. When standard L_2 weight decay is applied to W_Q and W_K separately, as is the default in essentially every modern optimizer, something quietly happens that nobody had cleanly articulated before: this routine penalty is no longer a routine penalty. It becomes mathematically equivalent to nuclear norm regularization on the product $W_Q W_K^T$, which is the convex relaxation of rank itself. In other words, the standard training recipe is implicitly biasing the attention weights toward low rank, with no one having explicitly asked for that.

The authors prove this in two steps. First, they show that if L_2 regularization were applied directly to a single matrix W , with an objective of the form

$$\mathcal{L}(W) = \mathcal{L}_{\text{data}}(W) + \frac{\lambda}{2} \|W\|_F^2, \quad (20)$$

This shrinks W uniformly and leaves its rank unchanged. The penalty has no opinion about rank in this form. Second, they show that the moment the same penalty is applied to a *factorized* parameterization $W = UV^T$ with $U \in \mathbb{R}^{m \times k}$ and $V \in \mathbb{R}^{n \times k}$, the objective

$$\begin{aligned} \mathcal{L}(U, V) = \mathcal{L}_{\text{data}}(UV^T) \\ + \frac{\lambda}{2} (\|U\|_F^2 + \|V\|_F^2) \end{aligned} \quad (21)$$

is provably equivalent to the nuclear norm formulation

$$\mathcal{L}_*(W) = \mathcal{L}_{\text{data}}(W) + \lambda \|W\|_*, \quad (22)$$

where $\|\cdot\|_*$ denotes the nuclear norm. The equivalence rests on the classical Srebro-Shraibman variational identity [2],

$$\|W\|_* = \min_{U, V: UV^T = W} \frac{1}{2} (\|U\|_F^2 + \|V\|_F^2), \quad (23)$$

which says the nuclear norm of W is exactly the minimum of the factored L_2 objective over all factorizations $W = UV^T$. Theorem 3.3 of [1] formalizes that the local minima of the factored objective coincide with those of the nuclear-norm one, and Theorem 3.4 strengthens this further by showing that during gradient flow training, the gap between the factored L_2 surrogate and the true nuclear norm decays exponentially fast. The decomposition itself, in short, is what creates the low-rank pressure—not the penalty form alone.

The natural next thought, and the one that motivates this section, was the following. The Burer-Monteiro factorization is not specific to transformers. It is the standard parameterization across an entire class of low-rank recovery problems: matrix completion, matrix sensing, low-rank matrix regression, recommender systems, and others. In all of these, the established regularizer is the same factored L_2 form $(\lambda/2)(\|U\|_F^2 + \|V\|_F^2)$. If the equivalence claimed by [1] is real, then in any of these problems we should be able to replace the factored L_2 penalty with a direct nuclear norm penalty $\lambda \|UV^T\|_*$ and recover the same solution. This section sets up that comparison on the matrix sensing problem and tests it empirically.

10.2 The Asymmetric BM Setting and the Question

The asymmetric Burer-Monteiro factorization $W = UV^T$ is the canonical parameterization in a wide class of low-rank recovery problems—matrix completion, matrix sensing, low-rank matrix regression, and inductive matrix completion among them. In nearly all such problems, the established regularizer is the factored L_2 form $(\lambda/2)(\|U\|_F^2 + \|V\|_F^2)$, chosen for its smoothness and separability over U and V . The nuclear norm $\|UV^T\|_*$, while theoretically the quantity of interest, is non-smooth and requires a singular value decomposition at each gradient step, making it computationally less attractive.

This raises a natural empirical question. If the factored L_2 formulation is mathematically equivalent to the nuclear norm formulation at the optimum—per the Srebro-Shraibman identity and Theorem 3.3 of [1]—do the two

penalties produce identical recovered matrices in practice? This section presents a controlled experiment to test this equivalence on the matrix sensing problem.

10.3 Problem Statement: Matrix Sensing

Matrix sensing is the canonical synthetic setting for low-rank recovery from linear measurements. An unknown matrix $M^* \in \mathbb{R}^{m \times n}$ of rank r^* is observed only through N scalar linear measurements

$$y_i = \langle A_i, M^* \rangle, \quad i = 1, 2, \dots, N, \quad (24)$$

where each $A_i \in \mathbb{R}^{m \times n}$ has entries drawn i.i.d. from $\mathcal{N}(0, 1/N)$. The Gaussian sensing ensemble satisfies the Restricted Isometry Property (RIP) with high probability when $N \gtrsim r^*(m+n)$. This condition guarantees the absence of spurious local minima in the BM landscape [3, 4].

The true minimum-rank-recovery formulation,

$$\min_W \text{rank}(W) \quad \text{subject to} \quad \langle A_i, W \rangle = y_i, \quad (25)$$

is NP-hard. The standard convex relaxation replaces the rank with the nuclear norm. Adopting the unconstrained Lagrangian formulation and applying the asymmetric BM factorization $W = UV^T$ yields the optimization actually solved in practice:

$$\begin{aligned} \min_{U, V} \quad & \frac{1}{2} \|\mathcal{A}(UV^T) - y\|_2^2 \\ & + \frac{\lambda}{2} (\|U\|_F^2 + \|V\|_F^2), \end{aligned} \quad (26)$$

where $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^N$ denotes the linear measurement operator with $[\mathcal{A}(W)]_i = \langle A_i, W \rangle$. Equation (26) is the standard formulation in the matrix-sensing literature.

10.3.1 The Two Variants Under Comparison

We compare Equation (26) against an alternative formulation in which the factored L_2 penalty is replaced by a direct nuclear norm penalty on the product:

$$\min_{U, V} \quad \frac{1}{2} \|\mathcal{A}(UV^T) - y\|_2^2 + \lambda \|UV^T\|_*. \quad (27)$$

The data fidelity term and the regularization strength λ are held identical. Only the penalty form differs.

10.4 Hypothesis

We hypothesize that the nuclear norm formulation, Equation (27), will recover a matrix of rank *less than or equal to* that recovered by the factored L_2 formulation, Equation (26). The intuition behind this is that the nuclear norm acts directly on the product UV^T , which is the quantity whose rank we care about, whereas the factored L_2 penalty acts on U and V separately and is only an upper bound on the nuclear norm during training—tight only at convergence. If the nuclear norm penalty is more aggressive or more direct in pruning singular values, this would show up as a strictly lower or equal recovered rank to the one of L_2 penalty. Theorem 3.3 of [1] predicts the two ranks should in fact be equal at convergence, which would mean the factored L_2 penalty is a tight surrogate, not just an upper bound. The experiment will tell us which is the case.

10.5 Methodology

10.5.1 Problem Generation

For each experimental cell, the ground truth matrix $M^* \in \mathbb{R}^{m \times n}$ was constructed as $M^* = U^*(V^*)^T$ with $U^* \in \mathbb{R}^{m \times r^*}$ and $V^* \in \mathbb{R}^{n \times r^*}$ having entries drawn i.i.d. from $\mathcal{N}(0, 1)$, then rescaled so that $\|M^*\|_F = 1$. This normalization keeps the matrix size constant across all values of r^* , so any differences observed across the rank sweep can be attributed to rank alone and not to changes in matrix scale.

The sensing operator was instantiated as a tensor of shape (N, m, n) with each A_i having entries i.i.d. from $\mathcal{N}(0, 1/N)$. The measurement vector $y \in \mathbb{R}^N$ was computed as $y_i = \langle A_i, M^* \rangle$ in the noiseless regime.

10.5.2 Optimization Setup

The BM factor matrices $U \in \mathbb{R}^{m \times k}$ and $V \in \mathbb{R}^{n \times k}$ were initialized with entries i.i.d. from $\mathcal{N}(0, 0.01^2)$. Optimization used the Adam optimizer with learning rate 10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and decoupled weight decay set to zero (the regularization is included explicitly in the objective). A maximum budget of 10,000 gradient steps was allowed per run, with early stopping triggered when the variation in the total loss over the trailing window of 100 steps fell below 10^{-8} .

10.5.3 Experimental Grid and Pairing

The first phase swept the true rank $r^* \in \{2, 3, 5, 8\}$. The remaining knobs were tied to r^* : the BM inner width $k = 2r^*$ (overcomplete) and the number of measurements

$N = 5r^*(m+n)$, well above the RIP threshold. The matrix dimensions were fixed at $(m, n) = (100, 100)$, the regularization strength at $\lambda = 0.1$, and the noise level at $\sigma = 0$ throughout. Three independent seeds were run per cell.

A critical design choice for the comparison was paired execution: within each (r^*, seed) cell, both the L_2 and the nuclear norm runs used the same M^* , the same sensing operator A , and the same initialization (U_0, V_0) . The only variable was the penalty form. This isolates the regularizer’s effect from sample-level randomness in the comparison.

10.5.4 Metrics

At the conclusion of each run, we computed the singular values of the recovered matrix $W = UV^T$ and reported two complementary rank estimates:

- **Effective Rank:** the number of singular values exceeding $0.01 \cdot \sigma_{\max}$, where $\sigma_{\max}$ is the largest singular value. This thresholded count is robust to floating-point noise floors.
- **Pseudo-Rank:** the smallest k such that the cumulative sum of the top- k singular values exceeds 95% of the total spectral mass.

10.6 Results and Discussion

The Phase 1 sweep produced 24 runs (4 values of $r^* \times 3$ seeds $\times 2$ penalties). Across all cells, the recovery succeeded and the comparison between the two penalty forms is decisive.

10.6.1 Rank Recovery and Penalty Equivalence

Figure 3 plots the pseudo-rank against the true rank r^* for both penalties. The two curves overlap exactly with the diagonal $y = r^*$ at every value of r^* tested. The effective rank, plotted on the same axes, is identical to the pseudo-rank in every cell—a consequence of the spectral cliff described below. The recovered rank lies far below the search-space ceiling $k = 2r^*$, confirming that the regularizer is performing genuine pruning rather than passively saturating the available BM width.

The singular value spectra (Figure 4) make the equivalence visually unambiguous. In every panel, the recovered matrix exhibits a sharp cliff at index r^* , with kept singular values of magnitude $\mathcal{O}(0.1)$ to $\mathcal{O}(1)$ and pruned singular values sitting at the optimizer’s noise floor of approximately 10^{-7} . The drop spans six to seven orders of magni-

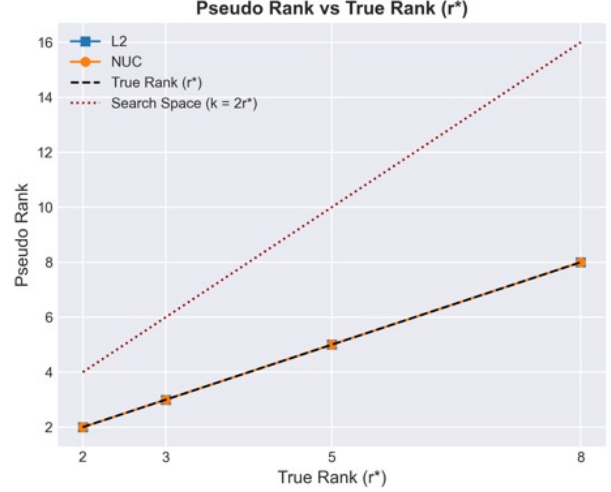


Figure 3. Both penalties recover rank exactly equal to r^* across all tested values, with the search-space ceiling $k = 2r^*$ shown as reference.

tude, leaving no ambiguity about the recovered rank. The L_2 and nuclear norm spectra coincide throughout—kept singular values, noise floor, and the slope of the spectrum within the kept block are all visually indistinguishable. The two penalties do not merely produce the same rank; they produce essentially the same matrix W .

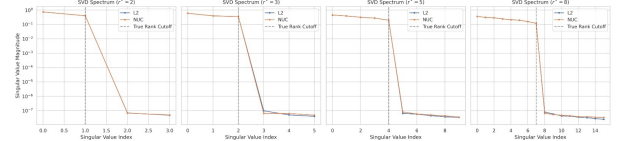


Figure 4. Singular values of $W = UV^T$ at convergence, log scale. The cliff at index r^* separates kept singular values ($\mathcal{O}(0.1)$ to $\mathcal{O}(1)$) from pruned ones ($\mathcal{O}(10^{-7})$). L_2 and nuclear norm curves overlap.

These results confirm Hypothesis H1: the factored L_2 and direct nuclear norm penalties on the asymmetric BM factorization produce identical recovered solutions in the matrix sensing setting. Theorem 3.3 of [1] is verified empirically across the rank sweep.

10.6.2 Iteration Cost: A Practical Asymmetry

While the two penalties agree at the optimum, they differ sharply in the number of gradient steps required to reach it. Figure 5 plots the iteration count against r^* for each penalty. The L_2 runs early-stop at approximately 1,500 steps for $r^* \in \{2, 3\}$ and around 4,000 steps for $r^* \in \{5, 8\}$. The nuclear norm runs, in contrast, exhaust the full 10,000-step budget in every cell.

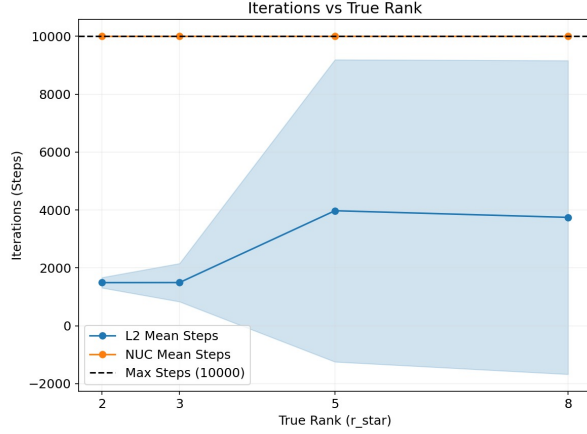


Figure 5. The L_2 runs converge under the early-stopping criterion well before the iteration budget is exhausted. The nuclear norm runs reach the maximum step count in every cell.

10.6.3 Why the Iteration Cost Diverges

The asymmetry has a clean explanation rooted in the structure of the two penalties:

- **The L_2 penalty is smooth and separable.** The term $(\lambda/2)(\|U\|_F^2 + \|V\|_F^2)$ decomposes elementwise over the entries of U and V , with gradient λU and λV respectively. Each gradient step is computationally trivial and well-conditioned, and the loss surface admits rapid convergence under Adam.
- **The nuclear norm penalty is non-smooth.** The term $\lambda \|UV^T\|_*$ requires a full singular value decomposition of the product UV^T at every gradient step in order to compute the penalty value and its gradient. The backward pass through the SVD involves operations on the full matrix of singular vectors, coupling all entries of U and V through a non-smooth function. Even after the rank has stabilized, the optimizer continues to register non-trivial gradient signal and does not satisfy the early-stopping criterion within the allotted budget.

This iteration-and-cost gap is precisely the practical reason the BM literature has uniformly settled on the factored L_2 formulation. The Srebro-Shraibman identity guarantees that L_2 on the factors is a faithful surrogate for nuclear norm regularization on the product, and the surrogate happens to be both smooth and decoupled. The non-smooth original is a worse computational choice for an identical statistical outcome.

10.7 Conclusion

This experiment provides a controlled empirical confirmation that, in the asymmetric Burer-Monteiro setting on a matrix sensing problem, the factored L_2 and direct nuclear norm regularizers converge to identical solutions. The recovered rank, the singular value spectrum, and the reconstruction error coincide between the two formulations across the swept range of true ranks. Theorem 3.3 of [1] is verified outside the transformer attention setting in which it was originally developed.

The experiment additionally surfaces a practical asymmetry: the nuclear norm formulation, while statistically equivalent, is substantially more expensive per gradient step due to the SVD it requires, and exhibits delayed convergence behavior under standard early stopping. This documents quantitatively the implicit cost-benefit calculation that underlies the dominance of the factored L_2 formulation in the matrix completion, matrix sensing, and recommender systems literature.

Subsequent phases of this study extend the rank sweep to $r^* \in \{10, 15, 20\}$, vary the regularization strength λ to map the rank-pruning transition, and investigate whether the equivalence persists under noise and incomplete observation regimes.

11. Discussion

11.1 Summary of the Research Journey

The path from classical BM theory to Margin Rank was non-linear. Table 2 summarises the key conceptual transitions.

Stage	Initial Assumption	What We Discovered
Classical BM	Pataki bound gives p^*	Bound inapplicable to asymmetric Softmax
Warp Drive	Softmax is just a nonlinearity	Softmax absorbs n constraints via null space
Implicit Reg	Nuclear norm oracle predicts p^*	Norm-dimension paradox; lazy optimizer
Margin Rank	Need SDP relaxation	Hard Bottleneck sweep is the correct oracle
GPT-2	All heads use similar capacity	Heterogeneous landscape; 80–97% redundancy

Table 2. Research journey: conceptual transitions and their causes.

11.2 Limitations and Caveats

- **Static attention.** The Hard Bottleneck Predictor requires a fixed, pre-computed attention matrix. It cannot be applied during training without significant overhead.
- **Margin sensitivity.** The choice of γ affects p^* ; the theory does not yet provide a principled method for selecting γ from first principles.

- **GPT-2 scale.** Our empirical profiling used GPT-2 (117M parameters). Whether the heterogeneous capacity landscape and 80% redundancy figures hold for frontier models (GPT-4, LLaMA 3) remains an open empirical question.
- **Conjecture 6.5.** The strict saddle property of the Softmax BM landscape at the corrected rank bound remains unproved.

12. Future Work

Several open problems emerge directly from this work:

1. **Completing the strict saddle proof.** The local surjectivity argument for the Softmax Jacobian restricted to the active constraint manifold is the key missing piece for a full proof of Conjecture 6.5.
2. **Non-commutative gradient flow.** Proving or disproving the nuclear norm implicit regularization conjecture in the non-commutative Softmax BM case.
3. **Frontier-model profiling.** Applying the Hard Bottleneck Predictor to LLaMA-3 and measuring whether Margin Rank scales sub-linearly with model depth.
4. **Training-time integration.** Designing a differentiable Margin Rank proxy that can regularize d_{head} during training, enabling Heterogeneous Transformer training from scratch.
5. **Precision-dependent formula fitting.** A formula

$$p^*(\varepsilon, \text{eff}_r, \tau, L) \quad (28)$$

incorporating precision, effective rank, temperature, and depth, fitted using `scipy.optimize.curve_fit` with confidence intervals.

13. Conclusion

This report has traced a complete research arc from classical Burer-Monteiro theory to a new framework—Margin Rank Theory—that mathematically characterises the true structural capacity of Transformer attention heads.

Our central results are:

1. The Barvinok-Pataki bound, while empirically verified in the symmetric SDP setting, is fundamentally inapplicable to Softmax attention due to the asymmetric geometry and the null space of the Softmax Jacobian.
2. The Softmax operator converts m linear equality constraints into margin inequalities, absorbing n row-sum constraints and enabling an order-of-magnitude dimensional compression (the Geometric Warp Drive: $p^* = 2$ for an Identity matrix of any size).
3. The Margin Rank of an attention head—the minimum embedding dimension required to geometrically separate targets from distractors—is the correct measure of head capacity, governed by topological complexity, not linear rank.
4. Real GPT-2 attention heads operate at 80–97% dimensional redundancy, with clear heterogeneous capacity requirements across heads in the same layer.

These findings provide rigorous mathematical justification for the development of **Heterogeneous Transformers**: future architectures where head dimensions are dynamically allocated by Margin Rank profiling, promising substantial reductions in computational footprint without sacrificing semantic expressivity.

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References

- [1] Kobayashi, S., Akram, Y., & von Oswald, J. (2024). Weight decay induces low-rank attention layers. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2410.23819.
- [2] Srebro, N., & Shraibman, A. (2005). Rank, trace-norm and max-norm. *Conference on Learning Theory (COLT)*, 545-560.
- [3] Recht, B., Fazel, M., & Parrilo, P. A. (2010). Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization. *SIAM Review*, 52(3), 471-501.
- [4] Tu, S., Boumal, N., Bhojanapalli, S., Sanghavi, S., & Recht, B. (2016). Low-rank solutions of linear matrix equations via Procrustes flow. *International Conference on Machine Learning (ICML)*.

- [5] Burer, S., & Monteiro, R. D. (2003). A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization. *Mathematical Programming*, 95(2), 329–357.
- [6] Burer, S., & Monteiro, R. D. C. (2003). A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization. *Mathematical Programming*, 95(2), 329–357.
- [7] Burer, S., & Monteiro, R. D. C. (2005). Local minima and convergence in low-rank semidefinite programming. *Mathematical Programming*, 103(3), 427–444.
- [8] Boumal, N., Voroninski, V., & Bandeira, A. S. (2016). The non-convex Burer-Monteiro approach works on smooth semidefinite programs. *Advances in Neural Information Processing Systems*, 29.
- [9] Gunasekar, S., Woodworth, B., Bhojanapalli, S., Neyshabur, B., & Srebro, N. (2017). Implicit regularization in matrix factorization. *Advances in Neural Information Processing Systems*, 30.
- [10] Zhu, M., Hayashi, H., & Ma, H. (2023). Scaling convex neural networks with Burer-Monteiro factorization. *International Conference on Learning Representations*.
- [11] Lee, J. D., Simchowitz, M., Jordan, M. I., & Recht, B. (2016). Gradient descent only converges to minimizers. *Conference on Learning Theory*, 1246–1257.
- [12] Candès, E. J., & Recht, B. (2009). Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6), 717–772.
- [13] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- [14] Srebro, N., & Shraibman, A. (2005). Rank, trace-norm and max-norm. *Conference on Learning Theory*, 545–560.
- [15] Kobayashi, S., Akram, Y., & von Oswald, J. (2024). Weight decay induces low-rank attention layers. *Advances in Neural Information Processing Systems*.
- [16] Arora, S., Cohen, N., Hu, W., & Luo, Y. (2019). Implicit regularization in deep matrix factorization. *Advances in Neural Information Processing Systems*, 32.
- [17] Barvinok, A. (1995). Problems of distance geometry and convex properties of quadratic maps. *Discrete & Computational Geometry*, 13, 189–202.
- [18] Pataki, G. (1998). On the rank of extreme matrices in semidefinite programs and the multiplicity of optimal eigenvalues. *Mathematics of Operations Research*, 23(2), 339–358.
- [19] Waldspurger, I., & Waters, A. (2020). Rank optimality for the Burer-Monteiro factorization. *SIAM Journal on Optimization*, 30(3), 2577–2602.